Theory of Computing
Turing Machine - 1

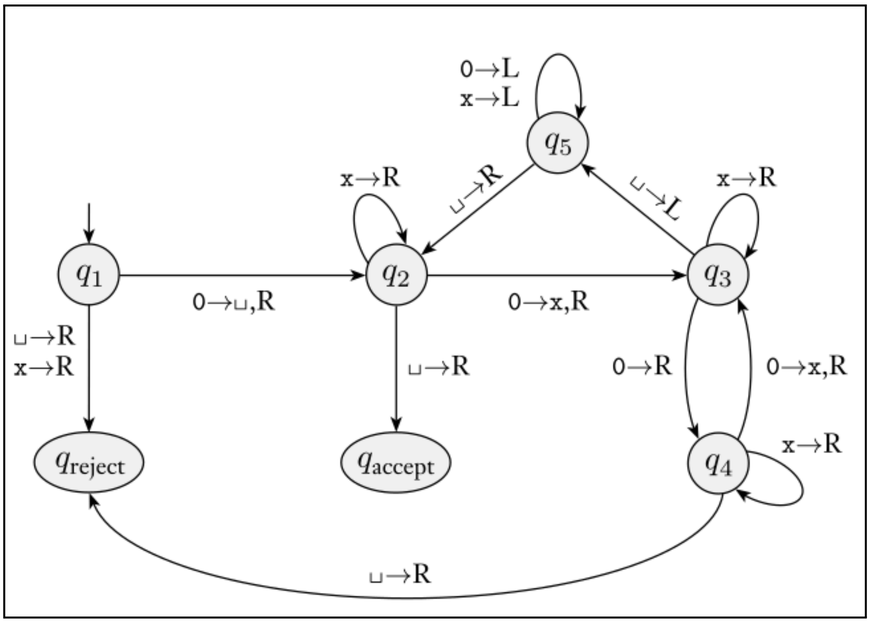
TD 10/W12

2ND YEAR - ENSIA

PRE-TUTORIAL EXERCISES

Given the following state diagram for the Turing Machine.

1. Run a simulation for each of the following strings giving the configuration at each step and deduce whether the word is accepted or not.
    a. 0.
    b. 00.
    c. 000.
    d. 000000.
2. What's the Language described by this Turing Machine?



EXERCISES

Exercise C1 (Creating TM):

Let the language L = { a^i b^j c^k } such that i = j + k.

- Write in **plain english,** the algorithm.
- Draw the Turing machine which recognizes L. (**NO USE OF PC**)

Exercise C2 (Creating a TM)

Draw the Turing Machine diagram for Even-Length Palindromes. (**NO USE OF PC**)

Exercise C3 (Creating TM):

Design a Turing Machine to calculate the parity of a binary number, i.e. add a 0 at the end if the number of 1's in the input string is even or a 1 if this number is odd.

Exercise P1 (Optional)

Design a Turing Machine to be a unary counter of characters in the language with alphabet Σ = {a,b,c}, i.e., the machine must generate as 1's as output as characters in the input word.

Exercise P2 (Optional)

Give implementation-level descriptions of Turing machines that decide the following languages over the alphabet {0,1}.

1. {w| w contains an equal number of 0s and 1s}
2. {w| w contains twice as many 0s as 1s}
3. {w| w does not contain twice as many 0s as 1s}

Exercise P3 (Optional)

For each of the following languages, draw a transition diagram for a Turing machine that accepts that language.

1. $\{a^i b^j \mid i < j\}$
2. $\{a^i b^j \mid i \leq j\}$

Exercise P4 (Optional)

Design a Turing Machine which takes two input words generated with the alphabet {0,1,2}, separated using the symbol {#}, and verifies whether they are the same. For instance, given the input b2101#2101b the Turing Machine would inform that both words are the same, where b represents empty cells in the tape.

Exercise P5 (Optional) [Recommended]

Design a Turing machine which automates verifying whether a **given** word is recognized by a **given** DFA.

Exercise P6 (Optional)

For each part below, draw a transition diagram for a TM that accepts AEqB = $\{x \in \{a, b\}^* \mid n_a(x) = n_b(x)\}$ by using the approach that is described.

1. Search the string left-to-right for an a; as soon as one is found, replace it by X, return to the left end, and search for b; replace it by X; return to the left end and repeat these steps until one of the two searches is unsuccessful.
2. Begin at the left and search for either an a or a b; when one is found, replace it by X and continue to the right searching for the opposite symbol; when it is found, replace it by X and move back to the left end; repeat these steps until one of the two searches is unsuccessful.

Exercise P7 (Optional)

Design a Turing Machine to generate a copy of a string with symbols {A,B,C}. For instance, given the input “bAABCAb”, the resulting input tape would be “bAABCAAABCAb”, where b represents the blank symbol.

Exercise P8 (Optional)

Design a TM to recognize the language of strings of the form $0^n 1^n 2^n$.